

Luis Lara

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EDUCATION

University of California, Berkeley

Berkeley, CA

B.S in Mechanical Engineering, Minor in EECS

Expected Grad: Dec 2028

Relevant Coursework: Solid Mechanics, Python, MATLAB, C++, Electricity and Magnetism, Circuits

Regents' and Chancellor's Scholar: UC Berkeley's merit-based award, granted to the **top 1%** of entering students

WORK EXPERIENCE

Lawrence Berkeley National Laboratory

Berkeley, CA

Biomechanics Research Intern

Jun 2025 – Aug 2025

- **Conducted** biomechanics data analysis to study **10+ high-speed** motion of sharpshooter insects, generating insights relevant to bio-inspired robotic mechanisms for researchers at **Georgia Tech Robotics**.
- Developed **10+** detailed 3D models of sharpshooter insects using **MeshLab, Fusion 360, and Blender** for **10000+** scans
- **Led** a team of **3** in developing a **VR** visualization system that allowed better analysis of **10+** 3D bug models.
- **Conducted** experiments at **ALS Beamline 8.3.2** to collect high-resolution scan data for biomechanics analysis for **50+** bugs.
- **Presented** research to **200+ researchers** at conference and **50+ scientist** at the ALS building

Lawrence Berkeley National Laboratory

Berkeley, CA

BLDAP (Berkeley Lab Directors Apprenticeship Program)

Jun 2024 – May 2025

- Used **NumPy, Pandas, and Matplotlib** to **analyze, visualize and clean 15+** datasets across real research projects.
- **Programmed** Arduino and **10+** sensors and actuators in **C++** to support a **solar-powered, energy-efficient** house model.
- Designed **CAD** models in **Fusion 360** and applied **3D printing** to **manufacturer** components for an eco-efficient house
- **Taught** a class of **30+** elementary school students introductory coding with **Scratch** and 3D modeling with **Tinkercad**.

Experts Lumber Structures Corporation

Hayward, CA

Carpenter

Jun 2019 – Present

- Worked on **100+** large-scale construction and fabrication projects, including **houses, fences, barns, and concrete work**
- Supported project **planning and material management** for jobs with budgets up to **\$15,500+**.
- Performed **hands-on** construction work, including **12+ hours** in **fast-paced job-site environments**.
- **Extensive** experience operating power tools, including **drills, saws, nail guns, and concrete mixers**

Santa Clara University

Santa Clara, CA

SEEDs (Spring Engineering Education Days)

Mar 2024 - 2025

- Selected as **1 of 50 students from 500 applicants** to participate in free **hands-on** engineering projects.
- Applied fundamentals of **aerodynamics** and **fluid mechanics** to **design 5+** models of **lightweight plastic frisbee**.
- Built an **Arduino** LED circuit, from fundamentals of **electrical circuits, wiring, and basic embedded programming**.
- Collaborated with a group to **design and construct** a bridge prototype capable of supporting a **75-lb payload**

ACTIVITIES/PROJECTS

Vehicle dynamics Project

- Developed a **MATLAB** vehicle handling model to predict steering angle vs. velocity and classify understeer, oversteer, and neutral-steer behavior for speeds up to **50 m/s**.
- Ran parameter sweeps across **10+** configurations of **weight distribution, center of mass, and tire stiffness**
- Generated **10+ data visualizations** analyzing handling response across varying loads, velocities, weight distributions
- **Debugged and validated** each function using test cases with known expected output

STAR (Space Technologies And Rocketry)

Sep 2025 – Dec 2025

Propulsion General Member

- Worked on the **combustion chamber and injector system** to support rocket propulsion performance, focusing on proper **geometry, fit, and manufacturability**.
- Considered key propulsion factors such as **fuel/oxidizer flow, injector spray behavior, pressure, heat, and chemical combustion** effects when supporting design decisions

Skills

Software: SolidWorks, Fusion 360, Onshape, Ansys, Python(Pandas, NumPy, C++, MATLAB, SQL,Git,GitHub,OpenCV,
Engineering: CAD, GD&T, FEA, soldering, 3D printing